

Chapter 7 Statistical Intervals Based on A Single Sample

A point estimation provides no information about precision and reliability of estimation. The point estimate says nothing about how close it might be to actual value.

So we use entire interval of plausible values, an interval estimate or confidence interval(CI). We want to find a interval(CI). The probability of the interval cover the true value is confidence level(CL).

7.1 Basic Properties of Confidence Intervals

Considering a Simple Case

- Population is normal distribution
- σ is known

■ Example 7.1

Industrial engineers who specialize in ergonomics are concerned with designing workspace and devices operated by workers so as to achieve high productivity and comfort. A sample of $n = 31$ trained typists was selected, and the preferred keyboard height was determined for each typist. The resulting sample average preferred height was 80.0 cm. Assuming that preferred height is normally distributed with $\sigma = 2.0$ cm. Please obtain a CI for μ , the true average preferred height for the population of all experienced typists.

Given the confidence level is 95%, find confidence intervals?

The sample mean is $\bar{X}$ and standard deviation is $\sigma/\sqrt{n}$.

Then $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$.

For confidence level $100(1 - \alpha)\%$, we find $P(-z_{\frac{\alpha}{2}} < Z < z_{\frac{\alpha}{2}}) = 1 - \alpha$.

i.e., $P\left(\bar{X} - \frac{\sigma}{\sqrt{n}}z_{\frac{\alpha}{2}} < \mu < \bar{X} + \frac{\sigma}{\sqrt{n}}z_{\frac{\alpha}{2}}\right) = 1 - \alpha$.

For $\alpha = 0.05$, i.e., the confidence level is 95%, corresponding $z_{0.025} = 1.96$.

Confidence Level, Precision, and Choice of Sample Size

Reliability is opposed to precision.

Higher confidence level, larger σ and smaller n correspond to wider interval.

7.2 Large-Sample Confidence Intervals for a Population Mean and Proportion

- Any population, normal or un-normal
- σ is known
- n is large enough ($n > 40$)

By CLT, $\bar{X}$ has approximately a normal distribution.

If σ is not known, we can use the S (variance) of sample to replace σ .

Proposition

If n is sufficiently large ($n > 40$), the standardized variable

$Z = \frac{\bar{X} - \mu}{S/\sqrt{n}}$ where S is the standard deviation of sample.

One-Sided Confidence Intervals

In some cases, we will want only the upper confidence or the lower one.

Proposition

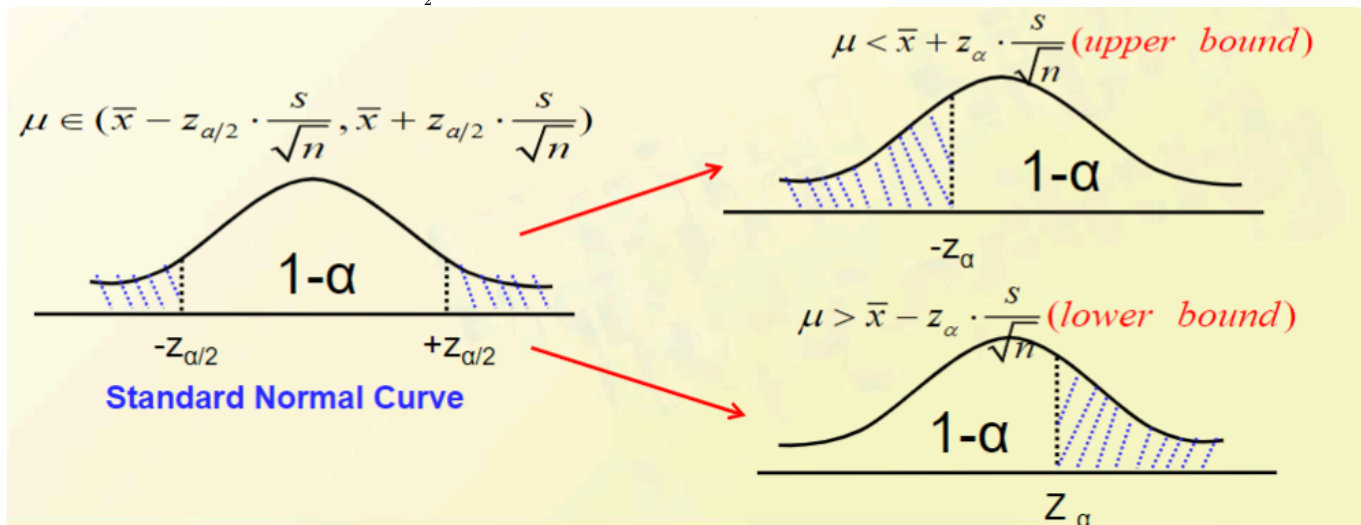
A large-sample upper confidence bound for μ is

$$\mu < \bar{x} + z_{\alpha} \cdot \frac{s}{\sqrt{n}}$$

and large-sample lower confidence bound for μ is

$$\mu > \bar{x} - z_{\alpha} \cdot \frac{s}{\sqrt{n}}$$

note that it is z_α rather than $z_{\frac{\alpha}{2}}$.



7.3 Intervals Based on a Normal Population Distribution

- Normal distribution
- μ and σ are unknown
- n is small

If n is small, the CLT cannot be invoked.

Theorem

The rv

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has a probability distribution called a **t distribution** with $n - 1$ degrees of freedom(df).

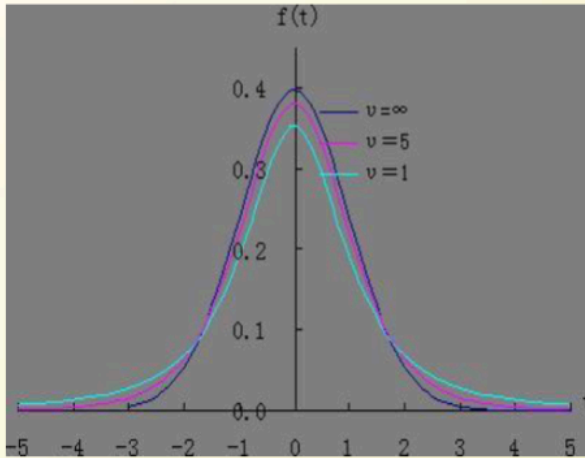
Properties of t Distribution

the only parameter in T is the number of df is $v = n - 1$.

$$T = \frac{\bar{X} - \mu}{S / \sqrt{n}}$$

The only one parameter in T is the number of df: $v=n-1$

Let t_v be the density function curve for v df



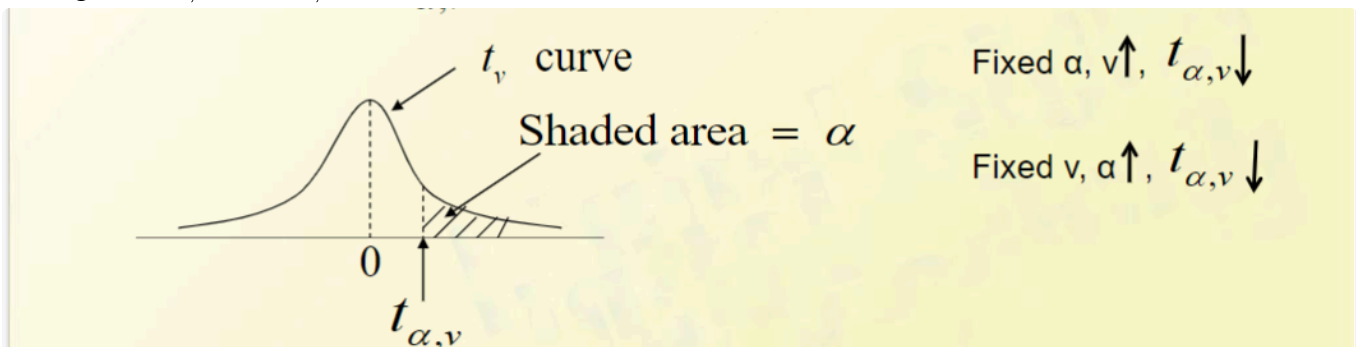
1. Each t_v curve is bell-shaped and centered at 0
2. Each t_v curve is more spread out than the standard normal curve.
3. As v increases, the spread of the corresponding t_v curve decreases.
4. As $v \rightarrow \infty$, the sequence of t_v curves approaches the standard normal curve $N(0,1)$.

Rule: $v \geq 40 \sim N(0,1)$

As $v \rightarrow \infty$, the sequence of t_v curves approaches the standard normal curve $N(0,1)$.

Notation

let $t_{\alpha,v}$ be the value on the measurement axis for which the area under the t curve with v df to the right of $t_{\alpha,v}$ is α ; $t_{\alpha,v}$ is called a t critical value.



Proposition

let $\bar{x}$ and s be the sample mean and sample standard deviation computed from the results of a random sample from a normal population with mean μ , then a $100(1 - \alpha)\%$ confidence interval for μ is

$$\left(\bar{x} - t_{\frac{\alpha}{2}, n-1} \cdot \frac{s}{\sqrt{n}}, \bar{x} + t_{\frac{\alpha}{2}, n-1} \cdot \frac{s}{\sqrt{n}} \right)$$